Sign Language Recognition & Translation Project Brief

Project Mission

The mission of this project is to design and develop a computer vision system capable of recognizing sign language gestures from video input and converting them into readable English text in real-time. This system aims to demonstrate the practical application of artificial intelligence in improving communication accessibility between deaf or hard-of-hearing individuals and the wider community.

Project Purpose

The purpose of this project is to build a functional prototype that showcases the integration of computer vision, machine learning, and real-time processing techniques. The system will serve as a proof-of-concept for sign language interpretation using affordable hardware such as a standard webcam.

Scope of the Prototype

This project focuses on isolated sign recognition using a limited vocabulary (20–50 signs). The system will process webcam input, extract hand and body landmarks, and classify gestures into corresponding English words.

Core Objectives

- Develop a real-time webcam-based sign recognition system - Implement landmark extraction using MediaPipe - Train a machine learning model for sign classification - Display predicted English text with confidence scores - Ensure system responsiveness and usability

Expected Outcome

A working prototype capable of recognizing selected sign language gestures in real-time and translating them into English words, demonstrated through an interactive webcam application.